

SUYASH SINGH

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EDUCATION

Manipal Institute of Technology

B.Tech in Computer Science Engineering (Data Science)

Expected Graduation: 2027

Bengaluru, India

- CGPA: 8.51/10 | Rigorous coursework in Data Structures & Algorithms, Operating Systems, and Linear Algebra.

TECHNICAL SKILLS

Languages & Core: C++ (C++20), Python, SQL, C, Linux/Unix, Bash, Data Structures & Algorithms, Operating Systems

Quantitative & ML: Market Microstructure, Order Flow Dynamics, Time-Series Modeling, LightGBM, PyTorch, HDF5, NumPy

Low-Latency Systems: Memory Pool Allocators, Cache Profiling (perf), Branch Prediction, GTest, Concurrency, Multithreading

Platforms & Infra: Docker, Kubernetes, Redis, PostgreSQL, Prometheus, Grafana, AWS, Git, CI/CD pipelines

Competitive Programming: LeetCode (200+ Problems Solved) | CodeChef 3★ (Algorithmic Problem Solving & Speed)

TECHNICAL PROJECTS

Low-Latency Limit Order Book Matching Engine

C++20, STL, GTest, Linux perf

- Exchange Simulation:** Engineered a deterministic matching engine executing limit/market orders under sub-microsecond latencies using hash maps and doubly linked lists for $O(1)$ FIFO price-time priority updates.
- Memory Optimization:** Integrated a custom fixed-size memory pool allocator to eliminate runtime heap allocation (malloc/new) overhead, mitigating memory fragmentation and reducing P99 tail latency by 65%.
- Hardware Hotspots:** Profiled kernel bottlenecks with Linux perf; minimized branch mispredictions via `[[likely]]/[[unlikely]]` annotations, cutting CPU instruction cache misses by 40%.

High-Frequency Order Flow Imbalance (OFI) Predictor

Python, NumPy, Pandas, LightGBM, HDF5

- Tick-Level Pipeline:** Engineered an end-to-end analytical pipeline parsing 50M+ millisecond-level tick records from HDF5 to model order book dynamics and forecast mid-price movements across 5–50 tick horizons.
- Microstructure Alpha:** Formulated microstructure features (Order Flow Imbalance, VPIN, Spread Resilience), achieving a 12% predictive accuracy gain over baseline momentum signals.
- Zero-Leakage Testing:** Implemented walk-forward cross-validation to eliminate lookahead bias and guard against time-series overfitting under real-world trading constraints.

Self-Erasing Neural Networks (SENNs) — Research Publication, ICDDs 2025

Python, PyTorch, Data Structures & Algorithms

- Mathematical Modeling:** Co-authored peer-reviewed algorithmic research on selective data erasure in deep networks, evaluating weight update dynamics and gradient shift trade-offs under rigorous constraints.
- Quantitative Evaluation:** Built diagnostic pipelines tracking class-level Hessian approximations, visualizing stability-plasticity trade-offs across targeted loss-gradient surfaces.

PathFlow — “Strava for AI Agents”

Next.js 15, TypeScript, Tailwind CSS, React Flow, Prisma, OpenTelemetry, Python

- Telemetry & Metrics:** Architected an OpenTelemetry-compatible telemetry platform monitoring runtime performance, token velocity (t/s), compute costs, and execution paths across asynchronous agent swarms.
- Execution Profiling:** Built an interactive DAG visualizer with React Flow to profile distributed trace execution trees, identifying execution bottlenecks and sub-span latencies in real time.

WORK EXPERIENCE

Omnisavant

Sept. 2026 – Present

Software Development Intern

Remote

- Engineered real-time orchestration pipelines and event-driven backend services, enabling low-latency state synchronization and interactive execution for autonomous demo engines.
- Optimized asynchronous data ingestion and streaming telemetry across concurrent client sessions, improving backend throughput and eliminating runtime latency bottlenecks.

Turing

Dec. 2025 – May 2026

AI Intern

Bengaluru, India

- Engineered distributed, high-throughput asynchronous backend services on **AWS**, designing state machine transitions, concurrent webhook pipelines, and robust session recovery under heavy traffic.
- Implemented resilient routing mechanisms and fail-safe circuit breakers backed by automated testing suites, maintaining high service availability and predictable sub-millisecond execution loops.

POSITIONS OF RESPONSIBILITY

Codex (Competitive Programming Club)

2025

Project Head

- Conducted algorithmic problem-solving workshops and structured mock contests for **100+ members**, focusing on dynamic programming, graph traversal, and time/space complexity optimization.

Manipal Bengaluru Open-Source Community (MBOSC)

2024 – 2025

Project Head

- Mentored **200+ student developers** on software design, modular architectures, and rigorous code reviews, emphasizing production-grade clean code and version control best practices.